
HYBRID CHANCE-CONSTRAINED OPTIMAL POWER FLOW UNDER LOAD AND RENEWABLE GENERATION UNCERTAINTY USING ENHANCED MULTI-FIDELITY GRAPH NEURAL NETWORKS

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ABSTRACT

Power systems are transitioning towards renewable sources and electrification, introducing significant uncertainties in generation and demand that Optimal Power Flow (OPF) methods must manage. Traditional deterministic methods struggle with these variabilities. Additionally, addressing uncertainty in OPF calculations incurs computational burdens due to the need for multiple evaluations across various scenarios. This necessitates the use of advanced surrogate models. However, these models require significant data for training, and surrogate-based optimization can yield unreliable results due to inaccuracies in constraint handling. To overcome these issues, this paper proposes a novel surrogate-based Hybrid Chance-Constrained Optimal Power Flow (HCC-OPF) methodology employing Enhanced Multi-Fidelity Graph Neural Networks (EMF-GNN) as power flow solver surrogates. This model integrates low-fidelity and high-fidelity simulations to significantly reduce training cost while maintaining high accuracy. We further enhance the robustness and accuracy of OPF solutions through a hybrid methodology that selectively uses exact power flow solver to correct surrogate inaccuracies near critical thresholds. Extensive testing on multiple IEEE systems under high-dimensional correlated uncertainty of load and generation shows the EMF-GNN model outperforms existing single-fidelity and multi-fidelity models. Furthermore, the proposed HCC-OPF methodology accurately solves OPF problems across various system sizes and conditions, exhibiting scalability and efficiency. Additionally, it effectively manages N-1 security constraints to further exhibit its robustness under operational challenges.

Keywords Optimal Power Flow, Graph Neural Network, Multi-fidelity Data Fusion, Uncertainty, Renewable Energy Generation, Hybrid Optimization

1 Introduction

Optimal power flow problem (OPF) concerns determining the optimal operating levels of power system variables, specifically generator outputs, to minimize the generation cost while satisfying all operational constraints such as power balance, voltage limits, and transmission line capacities (1). With ever-growing electrification and integration of

renewable energy sources (2; 3; 4; 5) due to climate change effects (6), uncertainties in both the demand and generation patterns of power systems are increasing (7).

Robust optimization (8; 9) and *stochastic programming* (10; 11) are two widely used optimization frameworks selected for handling the associated uncertainties in OPF. However, robust optimization tends to yield overly conservative solutions to maintain stability and meet constraints under unlikely scenarios, often leading to underutilization of resources and increased operational costs. On the other hand, stochastic programming can be time-consuming due to the introduction of numerous additional variables and constraints (12). Recently, the *Chance Constrained Optimal Power Flow (CC-OPF)* (13; 14) has attracted attention for being less conservative and time-consuming. Specifically, CC-OPF simplifies the problem by incorporating probabilistic constraints, which only need to be satisfied with a certain probability level, rather than modeling every possible scenario explicitly. This approach, hence, provides a good balance between robustness and computational efficiency.

Still, CC-OPF requires numerous power flow (PF) analyses to evaluate the chance of violating each constraint, which limits its real-time applications given the computational cost of traditional power flow solver methods like Newton-Raphson (NR), especially in large systems. To mitigate this, many researchers have simplified the problem by using a direct current power-flow (DCPF) model (15; 16), which assumes constant voltage magnitudes at every bus (1 p.u.) and neglects reactive power. This simplification can mislead the analysis to sub-optimal solutions. Alternatively, others have further refined this approach by incorporating a first-order linearization of the AC power-flow model (17; 18). Although this method may perform well under the assumption of small deviations from operating point or steady-state analysis, it can provide inaccurate results when these assumptions are violated, potentially affecting the optimality or feasibility of solutions. A promising alternative to overcome these limitations is the development of advanced surrogate models that replace computationally expensive power flow models and enable fast probabilistic analysis (19). Machine learning methods such as Polynomial Chaos Expansion (PCE) (20; 21), and artificial neural networks (NNs) (22; 12) have been widely used as surrogates in power flow analysis due to their ability to handle non-linearity. However, these methods often lack the adaptability to power grids with different settings from those on which they were trained (23). To address these shortcomings, recent research has been exploring the use of Graph Neural Networks (GNNs) (24) as efficient surrogates for power flow analysis. GNNs are particularly effective for power systems due to their inherent graph structure, with nodes as buses and edges as transmission lines. GNN provides topology learning and spatial information understanding capabilities that allow robustness against sudden topology changes such as generators or line failures and planned re-routing activities (25). This makes GNNs suitable for handling the OPF N-1 security contingency constraints.

Similar to other machine-learning surrogates, training accurate GNNs depends on the availability of a large amount of high-fidelity training data (26; 27). However, obtaining such a large number of training simulations can be computationally prohibitive, especially for large systems (28). To address this challenge, we adopt multi-fidelity modeling which has proven its efficiency in reducing training costs in our previous study (29). Specifically, this multi-fidelity GNN (MF-GNN) surrogate model fuses computationally cheap low-fidelity and computationally expensive high-fidelity power flow simulations to achieve high accuracy with significantly reduced computational costs by leveraging the precision of high-fidelity data alongside the abundance of low-fidelity data. Building on this groundwork, we propose an Enhanced multi-fidelity GNN model (EMF-GNN) that improves the architecture of the previous model to achieve even better accuracy.

Leveraging surrogates within the CC-OPF framework introduces additional challenges that are mostly overlooked in the literature. Specifically, the error introduced by the surrogate PF models, although considered insignificant for standalone PF solutions, can have disproportionately large effects when used under OPF frameworks, with the possibility of critical operational constraints violations. Such violations can skew the entire optimization process, resulting in solutions that are not only suboptimal but potentially unreliable as they may exceed the desired chance constraints. To tackle this problem, a hybrid CC-OPF (HCC-OPF) approach is proposed that switches to the deterministic alternating current power flow (ACPF) solver only for those instances where it is not clear if the chance constraints are violated due to the error in the surrogate model.

We demonstrate the efficiency of our proposed CC-OPF method using EMF-GNNs across various IEEE standard cases (30). First, we compare the accuracy of our EMF-GNN surrogate model with MF-GNN and a single-fidelity model (SF-GNN) trained solely on high-fidelity simulations. Next, we assess the efficiency of our proposed method in handling operational constraints and N-1 security constraints.

The main contributions of this paper are summarized as follows:

- The proposed EMF-GNN enhances the multi-fidelity GNN-based power flow calculation surrogate in (29) in terms of accuracy and computational efficiency.

- An HCC-OPF framework has been proposed for OPF under uncertainty, which leverages EMF-GNN as the power flow solver surrogate.
 - HCC-OPF ensures precise constraint satisfaction and optimal solutions despite surrogate estimation errors.
 - Using GNNs as surrogates in HCC-OPF provides robustness against N-1 contingencies, characterized by a single generator or branch failure.

The rest of the paper is structured as follows. Related works and Background are presented in Section 2 and Section 3, respectively. The proposed framework is outlined in Section 4. Experimental results are discussed in Section 5 and concluding remarks are provided in Section 6.

2 Related Works

In recent years, surrogate models for PF and surrogate-based OPF have become popular (31; 32) to speed up their computation time under uncertainty. Initial surrogate modeling mostly involved non-linear regression methods such as Partial Least Squares, Gaussian Process Regression, and lifting dimensions (33; 34; 35). However, these models are often seen as too simplistic for capturing the complex dynamics in power systems. As a solution, different variations of deep learning models, including deep neural networks (DNNs) (36), radial basis function neural networks (RBFNNs) (37), physics-guided deep neural networks (38), and conditional generative adversarial networks (CGANs) (39), have been proposed as alternatives. GNN-based models, such as Graph Convolutional Networks (GCNs) (40) further improve the robustness and scalability of the power flow surrogates leveraging the graph structures in the electric grids. Moreover, incorporating physical laws directly into the GNN architecture has further enhanced the accuracy and robustness of these models (41). Recent advancements also include combining Graph Attention Networks (GATs) with physics-informed approaches to focus on the most relevant parts of the data (42; 43). Additionally, Typed Graph Neural Networks (T-GNNs) were introduced to handle different types of nodes and edges in the power system graph, enabling more nuanced and context-aware analysis (23). However, the main limitation of these models lies in the requirement for large volumes of high-fidelity data to train the models.

Compared to surrogate-based PF analysis, fewer works have focused on surrogate-based OPF under uncertainty. In fact, the majority of works concerning surrogates in OPF do not consider any source of uncertainty (44; 45; 46; 25; 47; 48). Among those that consider uncertainty, few studies have employed deep neural networks to tackle real-time OPF under uncertainty in a distribution network (22). Deep Quantile Regression-based surrogate models have also been utilized to handle joint chance constraints, predicting quantiles of maximum constraint violation and power loss in (12). GNNs have also been used by (49) in the context of OPF under uncertainty. However, the authors have leveraged GNN models to predict the system's power balance constraint, rather than a power flow solver. Consequently, the literature lacks investigation of the potential of GNN surrogate models for estimating power system states while solving OPF under uncertainty. Most importantly, existing approaches fail to adequately address the risk of constraint violations caused by inaccuracies in surrogate estimations. Given these challenges, hybrid surrogate-based optimization using the capabilities of both GNN models and the AC power flow solvers will embark as a bridge for these existing gaps.

3 Background

3.1 Optimal power flow problem

To explain the CC-OPF formulation, it is essential to first review the background on *the power flow analysis problem* (50; 51). Power flow analysis involves the computation of bus voltage phasors (magnitude and angle) given the generation and load demands for an electrical grid, which consequently provides the injections and flows of currents and real and reactive powers throughout the grid. The power flow problem is usually solved using the AC power flow model, which uses the following non-linear equations to describe the relationship between the electrical quantities:

$$P_i = |V_i| \sum_j |V_j| (G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j)), \quad (1)$$

$$Q_i = |V_i| \sum_j |V_j| (G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j)), \quad (2)$$

where P_i and Q_i are the real and reactive power injections at bus i , $|V_i|$ and θ_i represent the voltage magnitude and phase angle at bus i , and G_{ij} and B_{ij} are the conductance and susceptance of the line connecting buses i and j . These non-linear equations are typically solved iteratively using methods such as Newton-Raphson (52) or Gauss-Seidel (53).

Due to the computational intensity of solving these non-linear equations for large-scale grids, the DC power flow model offers a simplified linear approach that focuses primarily on the active power components and assumes a fixed voltage magnitude (1 per unit) across the network (54):

$$P_i = \sum_j B_{ij}(\theta_i - \theta_j). \quad (3)$$

The computational efficiency of DCPF makes it valuable for initial analyses and scenarios where a rapid approximation of grid behavior is necessary.

Moving forward to the OPF problem, this problem is focused on identifying cost-minimizing generation outputs that adhere to physical and operational constraints of the power grid (55). OPF seeks to minimize a cost function related to the real power generation from each traditional (fossil fuel) power plant as the decision variables. The mathematical formulation of this objective is given by:

$$\min_{P_{gi}} C = \min_{P_{gi}} \sum_{i=1}^n c_i(P_{gi}) = \min_{P_{gi}} \sum_{i=1}^n (a_i P_{gi}^2 + b_i P_{gi} + c_i), \quad (4)$$

where C denotes the total generation cost, P_{gi} denotes the power generation at bus i , and a_i , b_i , and c_i are cost coefficients that commonly form a quadratic cost function (56). The optimization is subject to operational constraints including power balance, voltage limits, generation limits, branch flow limits, and voltage angle differences:

$$\begin{aligned} \sum_{i=1}^n P_{gi} - \sum_{i=1}^n P_{di} - \sum_{i,j} P_{loss_{ij}} &= 0, \quad (\text{Active Power Balance}) \\ V_i^{min} &\leq V_i \leq V_i^{max}, \quad (\text{Voltage Magnitude at bus } i) \\ P_{gi}^{min} &\leq P_{gi} \leq P_{gi}^{max}, \quad (\text{Generation Limits at generator } i) \\ 0 &\leq |P_{ij}| \leq P_{ij}^{max}, \quad (\text{Branch Flow Limits for line } ij) \\ |\theta_i - \theta_j| &\leq \theta_{ij}^{max}, \quad (\text{Voltage Angle Difference between buses } i \text{ and } j) \end{aligned} \quad (5)$$

where P_{di} represents the real power demand at bus i , P_{ij} indicates the real power flow from bus i to bus j , $P_{loss_{ij}}$ denotes the power loss from bus i to bus j , V_i is the voltage magnitude at bus i , and θ_i and θ_j are the voltage angles at buses i and j , respectively. Parameters V_i^{min} and V_i^{max} set the voltage limits, and P_{gi}^{min} and P_{gi}^{max} set the generation limits. P_{ij}^{max} is the maximum allowable power flow on the branch connecting buses i and j , and θ_{ij}^{max} is the maximum allowable voltage angle difference between connected buses, i and j .

3.2 Chance-constrained optimal power flow formulation

The CC-OPF solves the OPF under uncertainty problem by introducing probability constraints that ensure inequality operational constraints are met with a specified probability, providing probabilistic assurance against violations under uncertain conditions (57). For power balance, which is typically modeled as an equality constraint, a single set of decision variables may not suffice to satisfy the balance under various realizations of demand and renewable generation uncertainties. Therefore, operational reserves are relied upon to manage these discrepancies. According to industry standards, reserves typically include approximately 6% for contingency (58; 59; 60), 1% for regulation (61), and 5% for ramping (62), summing up to around 10% of the total generation capacity available as a reserve. Consequently, the power balance equality constraints are revised to inequality constraints, permitting a 10% relaxation specifically in the direction of over-generation. Under-generation incurs penalties that are added to the objective function to reflect its economic impact. Additionally, any over-generation is also penalized to account for the costs associated with electricity spillage. Thus, the objective function is designed to minimize the expected generation costs while incorporating penalties for both under-generation and over-generation. The mathematical formulation of this optimization problem is

given by:

$$\min_{P_{gi}} \bar{C} = \min_{P_{gi}} \mathbb{E} \left[\sum_{i=1}^n c_i(P_{gi}) + c_{i,\text{under}}(\max(0, \sum_{i=1}^n P_{d_i} - \sum_{i=1}^n P_{g_i})) + \sum_{i=1}^n c_{i,\text{over}}(\max(0, P_{g_i} - P_{d_i})) \right] \quad (6)$$

s.t.

$$\begin{aligned} \mathbb{P} \left[\left(\sum_{i=1}^n P_{d_i} + \sum_{i,j} P_{loss_{ij}} - \sum_{i=1}^n P_{g_i} \right) \leq 0.1 \sum_{i=1}^n P_{g_i} \right] &\geq 1 - \epsilon, \\ \mathbb{P} \left[V_i^{\min} \leq V_i \leq V_i^{\max} \right] &\geq 1 - \epsilon, \\ \mathbb{P} \left[P_{g_i}^{\min} \leq P_{g_i} \leq P_{g_i}^{\max} \right] &\geq 1 - \epsilon, \\ \mathbb{P} \left[0 \leq |P_{ij}| \leq P_{ij}^{\max} \right] &\geq 1 - \epsilon, \\ \mathbb{P} \left[|\theta_i - \theta_j| \leq \theta_{ij}^{\max} \right] &\geq 1 - \epsilon \end{aligned} \quad (7)$$

where $\mathbb{E}$ denotes the expectation operator, $\bar{C}$ is the mean total generation cost, and ϵ is a small, acceptable risk level, reflecting the probability of constraint violations. While the value of ϵ is typically set to a value around 0.05 in the literature (63; 14; 13; 64), it is flexible to adapt to specific operational needs. Higher risk levels may reduce system costs in the short term but increase the likelihood of constraint violations in the long term. Such violations can lead to severe issues like system instability, voltage collapse, and increased incidence of power outages or component damages. Moreover, considering the relative importance and critically among the constraints from the system stability perspective, the proposed algorithm allows for setting different ϵ values for different constraints.

It is noteworthy that to enhance system reliability, OPF solvers must also account for N-1 contingency scenarios, ensuring the grid remains stable and operational even if a single component fails (65). This requires re-evaluating the system constraints under each potential single failure, including line or generator failure, to confirm the integrity and safety of the power network.

3.3 Graph Neural Networks (GNN)

GNNs are a class of deep learning models designed to perform inference on data described by graphs. The functionality of GNNs centers around the message-passing mechanism, where nodes aggregate information from their neighborhood to compute new feature representations. This mechanism can be described in two main phases: *message aggregation* and *node update*. In the message aggregation phase, each node collects information from its immediate neighbors to construct a comprehensive feature representation. Mathematically, the message aggregation at a node v as defined by Scarselli et al. (24) is defined as:

$$m_v^{(t)} = \sum_{u \in \mathcal{N}(v)} f(h_u^{(t-1)}, h_v^{(t-1)}, e_{uv}), \quad (8)$$

where $m_v^{(t)}$ represents the aggregated message for node v at layer t , $h_u^{(t-1)}$ and $h_v^{(t-1)}$ are the features of the neighbor node u and node v from the previous layer, respectively, and e_{uv} denotes the edge features, and f is a learnable function specific to the GNN architecture. Node features are updated by integrating the aggregated messages with their current states in the subsequent phase:

$$h_v^{(t)} = g(h_v^{(t-1)}, m_v^{(t)}), \quad (9)$$

where $h_v^{(t)}$ is the updated feature of node v at layer t , and g is another learnable function that combines the old features with the new messages. Messages passing over multiple layers provide each node with information from further nodes.

k-GNNs (GraphConv) (66) and Topology adaptive graph convolutional network (TAGConv) (67) are two extensions of the basic GNN structure described above, used in high-fidelity and low-fidelity blocks of the proposed multi-fidelity GNN structure in this paper, after various GNN operators including GAT (68), EdgeConv (69), GraphSAGE (70), SGConv (69), APPNP (71), ChebConv (72), AGNNConv (73), GCNConv (74), TagConv, and k-GNNs are tested for both blocks.

k-GNN refines the message aggregation mechanism by incorporating both node and edge features more explicitly using the following message-passing expression:

$$h_v^{(t)} = \sigma \left(W^{(t)} \sum_{u \in \mathcal{N}(v)} \theta_{uv} h_u^{(t-1)} + b \right), \quad (10)$$

where $W^{(t)}$ represents a weight matrix at layer t , θ_{uv} denotes the weight or strength of the edge between nodes u and v , and b is the bias term. The non-linear function, σ (such as ReLU) enables capturing complex data patterns.

TAGConv introduces a more flexible approach to message aggregation by accounting for varying neighborhood sizes and structures using the following message-passing expression:

$$h_v^{(t)} = \sigma \left(\sum_{k=0}^K \sum_{u \in \mathcal{N}_k(v)} W_k h_u^{(t-1)} \right), \quad (11)$$

where $\mathcal{N}_k(v)$ denotes the set of k -hop neighboring nodes of node v , allowing the model to consider both local and more distant neighborhood information. W_k are distinct weight matrices for different hop distances k , and σ has a similar role as in k-GNN.

4 Methodology

This section presents our proposed methodology for addressing the OPF problem under uncertainty using a novel hybrid CC-OPF method leveraging multi-fidelity GNN models. We begin by describing how we represent the power grid as a graph, with buses as nodes and transmission lines as edges. Following this, we detail our EMF-GNN model, which serves as a surrogate for power flow calculations. Finally, we discuss how we integrate this surrogate model into our HCC-OPF framework, outlining the optimization method and its associated algorithm. The source code for the EMF-GNN model training and HCC-OPF methodology can be accessed through GitHub.

4.1 Graph-Based Power Grid Modeling

In this study, we model an electric power grid comprising N buses and L transmission lines as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. The set $\mathcal{V}$ represents the nodes of the graph, corresponding to the buses, and the set $\mathcal{E}$ comprises edges that represent the transmission lines connecting the buses. An edge e_{ij} is an element of $\mathcal{E}$ if there is a transmission line connecting bus i to bus j . We denote the number of nodes and edges by $|\mathcal{V}| = N$ and $|\mathcal{E}| = L$, respectively, where $|\cdot|$ indicates the cardinality of a set. The adjacency matrix for the graph $\mathcal{G}$ is defined as $\mathbf{A} = [a_{ij}]$, with $a_{ij} = 1$ if there is an edge, i.e., $e_{ij} \in \mathcal{E}$, and $a_{ij} = 0$ otherwise.

Each node $v_i \in \mathcal{V}$ is characterized by a node feature vector $\mathbf{x}_i$ that consists of the attributes of bus i . These attributes are defined as:

$$\mathbf{x}_i = [P_i^d, Q_i^d, P_i^g, Q_i^g, V_i^s], \quad (12)$$

where P_i^d , Q_i^d , P_i^g , and V_i^s respectively denote the active power demand, reactive power demand, generated active power, and voltage set-point at bus i . To model the distributed renewable energy generation into power grids, some load buses are randomly selected and assigned a random value of renewable generation to them. As a result, some load buses only have the load feature, while others have loads and renewable injection. In other words, load buses with no renewable generation typically have their generation-related elements (P_i^g , Q_i^g , and V_i^s) set to zero. While, for load buses with renewable generation, active power generation (P_i^g) is set to the renewable injection value. Additionally, for generation buses, the demand elements, P_i^d and Q_i^d , are set to zero. Our proposed method utilizes this node feature information within a GNN framework to estimate the bus voltage magnitude, $|V_i|$, at each load bus, and phase angle, θ_i , at each bus. We will elaborate on our GNN framework in the next section.

4.2 Multi-Fidelity Graph Neural Network Surrogate

In this work, we introduce an enhanced residual multi-fidelity GNN architecture, built on the MF-GNN introduced by our previous study (29) to efficiently compute bus voltage phasors under various operational scenarios. Figure 1 shows the schematic of the proposed EMF-GNN. As shown in the schematic, the EMF-GNN consists of two main components: the Low-Fidelity Graph Neural Network (LF-GNN) and the High-Fidelity Graph Neural Network (HF-GNN), which utilize data from DCPF and ACPF simulations, respectively. Compared to MF-GNN, EMF-GNN is improved by the extraction of an embedding output from the LF-GNN, and inputting it in the middle layers of the HF-GNN.

The main purpose of the LF-GNN is to predict low-fidelity estimations of the phase angles of the power system's buses using the data from DCPF analysis since DCPF considers a fixed value of 1.0 p.u. for voltage magnitude at all buses. Additionally, this network outputs an embedding from the layer right before the output layer, which is later used in the HF-GNN. The forward pass of this model is as:

$$\begin{aligned} H_{\text{LF}}^{(0)} &= \mathbf{x}, \\ H_{\text{LF}}^{(l)} &= \text{ReLU} \left(\text{GNN}_{\text{LF}}^{(l)} \left(H_{\text{LF}}^{(l-1)}, \mathbf{A} \right) \right), \quad l = 1, \dots, L-1, \\ \hat{\Theta}_{\text{LF}} &= \text{GNN}_{\text{LF}}^{(L)} \left(H_{\text{LF}}^{(L-1)}, \mathbf{A} \right), \\ \mathbf{E}_{\text{LF}} &= H_{\text{LF}}^{(L-1)}, \end{aligned} \quad (13)$$

where $\mathbf{x}$ represents the input feature matrix, $\mathbf{A}$ is the adjacency matrix of the power grid graph, $\text{GNN}_{\text{LF}}^{(l)}$ denotes the TagConv (67) layer at depth l , and ReLU refers to the Rectified Linear Unit activation function (75). $\hat{\Theta}_{\text{LF}}$ are the estimated low-fidelity phase angles, and $\mathbf{E}_{\text{LF}}$ is the embedding output from the penultimate layer of LF-GNN.

The HF-GNN utilizes GraphConv layers, also known as k-GNNs (66), to refine the LF-GNN voltage phasor estimations. The process begins with the HF-GNN receiving initial inputs comprised of the low-fidelity phase angles, $\hat{\Theta}_{\text{LF}}$, combined with a vector of unit magnitudes to represent the LF-GNN estimated voltage and the original node features. These are concatenated and fed into the first GraphConv layer. For subsequent layers, the methodology involves an iterative enhancement where the output from the previous layer is concatenated with the low-fidelity embedding, $\mathbf{E}_{\text{LF}}$, from the LF-GNN, which significantly aids in maintaining information throughout the deeper layers of the network. This input is processed by the next GraphConv layer, continuing this pattern until the final layer. By leveraging both the coarse approximations from the LF-GNN and the contextual embeddings at each step, the HF-GNN effectively refines its predictions and enhances the accuracy of the voltage magnitude and phase angle estimations. This model is designed to estimate the residuals between low-fidelity and high-fidelity estimations. The forward pass of the model is described in the following equations:

$$\begin{aligned} H_{\text{HF}}^{(0)} &= \text{Concat}(\mathbf{x}, \mathbf{1}, \hat{\Theta}_{\text{LF}}), \\ H_{\text{HF}}^{(1)} &= \text{ReLU}(\text{GNN}_{\text{HF}}^{(1)}(H_{\text{HF}}^{(0)}, \mathbf{A})), \\ H_{\text{HF}}^{(l)} &= \text{ReLU}(\text{GNN}_{\text{HF}}^{(l)}(\text{Concat}(H_{\text{HF}}^{(l-1)}, \mathbf{E}_{\text{LF}}), \mathbf{A})), \quad \text{for } l = 2, \dots, L-1, \\ \mathbf{R}_V, \mathbf{R}_\theta &= \text{GNN}_{\text{HF}}^{(L)}(\text{Concat}(H_{\text{HF}}^{(L-1)}, \mathbf{E}_{\text{LF}}), \mathbf{A}), \\ \hat{\mathbf{V}}_{\text{HF}} &= \mathbf{R}_V + \mathbf{1}, \\ \hat{\Theta}_{\text{HF}} &= \mathbf{R}_\theta + \hat{\Theta}_{\text{LF}}, \end{aligned} \quad (14)$$

where $\mathbf{1}$ represents the vector of ones, Concat refers to the concatenation operation that merges multiple feature vectors into a single vector, $\text{GNN}_{\text{HF}}^{(l)}$ denotes the GraphConv layer at depth l . $\mathbf{R}_V$ and $\mathbf{R}_\theta$ are the residuals added to the initial low-fidelity estimations to obtain the refined high-fidelity voltage magnitudes $\hat{\mathbf{V}}_{\text{HF}}$ and phase angles $\hat{\Theta}_{\text{HF}}$, respectively.

The loss function in the LF-GNN model is based on the mean squared error (MSE). This can be expressed as:

$$\mathcal{L}_{\text{LF}} = \frac{1}{T_l N} \sum_{j=1}^{T_l} \sum_{i=1}^N \left(\hat{\Theta}_{\text{LF},i}^{(j)} - \Theta_{\text{LF},i}^{(j)} \right)^2, \quad (15)$$

where $\hat{\Theta}_{\text{LF},i}^{(j)}$ and $\Theta_{\text{LF},i}^{(j)}$ represent the estimated and exact phase angles at bus i for the j -th training sample, respectively. N is the number of buses and T_l is the number of low-fidelity training data samples T_l .

The HF-GNN model employs a loss function over voltage magnitudes and phase angles using an MSE approach as well, expressed as

$$\mathcal{L}_{\text{HF}} = \frac{1}{T_h N} \sum_{j=1}^{T_h} \sum_{i=1}^N \left[\left(\hat{\mathbf{V}}_{\text{HF},i}^{(j)} - \mathbf{V}_{\text{HF},i}^{(j)} \right)^2 + \left(\hat{\Theta}_{\text{HF},i}^{(j)} - \Theta_{\text{HF},i}^{(j)} \right)^2 \right], \quad (16)$$

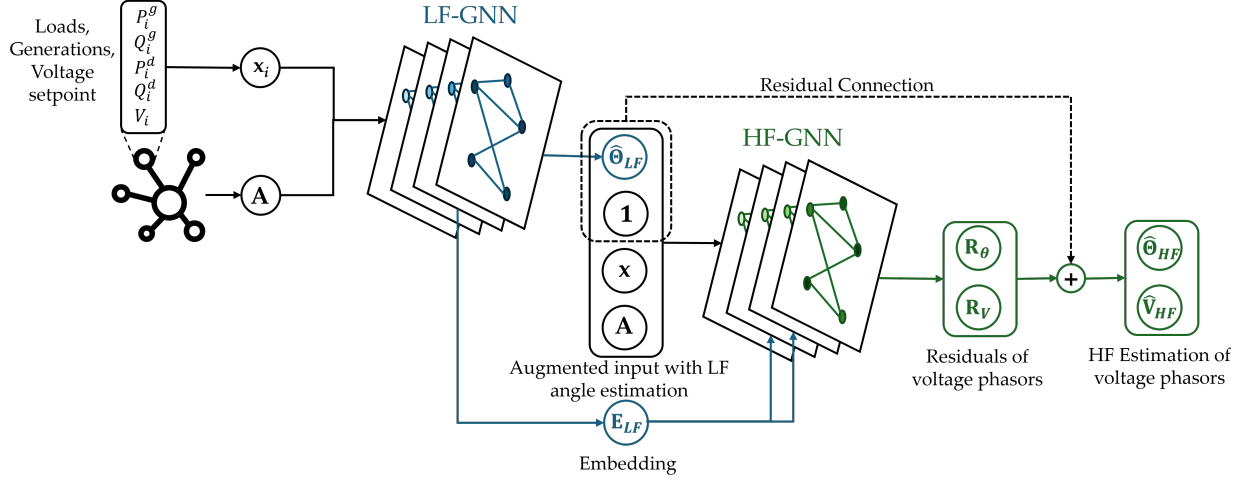


Figure 1: Architecture of the proposed EMF-GNN model, illustrating both the LF-GNN and HF-GNN components. The LF-GNN, using node features $\mathbf{x}$ and adjacency matrix $\mathbf{A}$, estimates low-fidelity phase angles, $\hat{\Theta}_{LF}$, and low-fidelity embeddings, $\mathbf{E}_{LF}$. The HF-GNN refines these estimations, producing high-fidelity estimates for voltage magnitudes, $\hat{\mathbf{V}}_{HF}$, and phase angles, $\hat{\Theta}_{HF}$, by estimating residuals $\mathbf{R}_\theta$ and $\mathbf{R}_V$, which are added to the initial LF-GNN outputs.

where T_h is the number of high-fidelity training data samples T_l . $\hat{\mathbf{V}}_{HF,i}^{(j)}$ and $\hat{\Theta}_{HF,i}^{(j)}$ denote the estimated voltage magnitude and phase angle at bus i for the j -th training sample, respectively; and $\mathbf{V}_{HF,i}^{(j)}$ and $\Theta_{HF,i}^{(j)}$ are their exact values obtained from ACPF.

4.3 Hybrid Chance-Constrained OPF

In this subsection, we present the proposed HCC-OPF under uncertainty methodology. It must be noted that to solve the OPF problem under high-dimensional uncertainty, we combined Chance Constrained Programming with Particle Swarm Optimization (PSO) (76). PSO, known for its simplicity and efficiency, handles uncertain variables and nonlinear spaces using a population-based approach. Particles move through the solution space, adjusting their paths based on their own experiences and the successes of their peers, enhancing the search for optimal solutions. This balance between exploration and exploitation makes PSO suitable for our task (77; 78).

The optimization begins with the initialization phase, where a population of potential solutions is initialized. The solutions (generator outputs) are repaired at this point to remain within generation limits. Optimization then proceeds with assessing the objective function value and constraint violations for each solution using a Monte Carlo (MC) sampling method. At this step, instead of running the ACPF for each realization of uncertainties, the EMF-GNN surrogate model is employed to estimate voltage phasors at each bus. Constraints are then evaluated using power system equations. Optimization then adds a penalty to the objective function value based on the extent of the violation to discourage the exploration of spaces that violate the constraints. However, when using a surrogate, even a highly accurate one, we cannot be certain about constraint violations, especially when the surrogate output is very close to the violation threshold.

To address this challenge, if the probability of constraint violation is close to the chance threshold ϵ , the proposed *hybrid approach* is activated. Specifically, the probability of constraint violation is recalculated by replacing the surrogate with exact ACPF for those MC samples that are within a predicated proximity of constraints. Optimization then proceeds with the updated constraint violation chances and updates the solutions. This process is repeated till convergence.

The pseudocode for the proposed HCC-OPF is presented in Algorithm 1, where N denotes the population size, M refers to the number of MC samples, K refers to the number of constraints, $\mathbf{RG}$ refers to the average renewable generation vector, $\bar{\mathbf{L}}$ represents the average load vector, $\bar{\mathbf{x}}_{dc}$ denotes the solution vector from the deterministic DC optimal power flow, $\mathbf{x}_i$ indicates the solution i in the population, $\mathbf{L}_j$, and $\mathbf{RG}_j$ are realizations of load and renewable generation vectors for each Monte Carlo sample respectively, Σ_L and Σ_{RG} denote the covariance metrics of load and renewable generation, $\mathbf{V}_j$ is the vector of voltage phasors computed using the surrogate solver in sample j , $C_{i,j}$ represents the cost associated with the i^{th} solution and j^{th} sample, $\mathbb{P}[g_k(\mathbf{V}_i, j)]$ denotes the probability of violating constraint k by

the i^{th} solution in sample j , δ refers to chance constraint proximity threshold, $\bar{C}_i$ denotes the mean generation cost for solution i , $\mathcal{P}_i$ refers to penalty cost for solution i , and S_i refers to the fitness score of solution i .

Algorithm 1: Hybrid Chance-Constrained Optimal Power Flow (HCC-OPF)

Data: Initialization data

Result: Optimal solution and objective function

Initialize:

for $i \leftarrow 1$ **to** N **do**

 Initialize solution $\mathbf{x}_i$ randomly

 Repair $\mathbf{x}_i$ to adhere to generation output limits

while *convergence not achieved* **do**

Evaluate Fitness:

for $i \leftarrow 1$ **to** N **do**

for $j \leftarrow 1$ **to** M **do**

 Draw samples $\mathbf{L}_j$ from $\mathcal{N}(\bar{\mathbf{L}}, \Sigma_L)$ and $\mathbf{RG}_j$ from $\mathcal{N}(\bar{\mathbf{RG}}, \Sigma_{RG})$

 Compute $\mathbf{V}_{i,j}$ using the EMF-GNN

 Compute $C_{i,j}$ and evaluate constraints

for $k \leftarrow 1$ **to** K **do**

 Compute $\mathbb{P}[g_k(\mathbf{V}_{i,j})]$

if $|\mathbb{P}[g(\mathbf{V}_{i,j})] - \epsilon| < \delta$ **then**

 Re-evaluate constraints using ACPF solver for critical samples

 Recompute $\mathbb{P}[g_k(\mathbf{V}_{i,j})]$

$\bar{C}_i = \sum_{j=1}^M C_{i,j} / M$

 Compute penalty cost $\mathcal{P}_i$

$S_i = \bar{C}_i + \mathcal{P}_i$

Update solutions based on $\{S_i\}$ for $i = 1, \dots$, population size

Check convergence criteria

return optimal solution and objective function

5 Results and Discussion

This section evaluates the merit of the proposed EMF-GNN and surrogate-based HCC-OPF in terms of accuracy and robustness. Specifically, we first investigate the improved accuracy of EMF-GNN and then proceed to evaluate its benefit when employed within the HCC-OPF framework in terms of closeness to optimal solution and robustness. Lastly, we examine the benefits of the proposed hybrid approach rather than fully relying on surrogates within CC-OPF.

5.1 Generation of Data for EMF-GNN Power Flow Model

All training, validation, and test data for the EMF-GNN model are generated using PyPower (Python equivalent of MATPower (79)) on standard IEEE test cases for consistency and reproducibility. The test cases used in this study are detailed in Table 1. The low-fidelity and high-fidelity data are generated using DCPF and ACPF features in PyPower. Each data sample is generated by randomly perturbing the loads and generator outputs from the default PyPower test cases to capture the realistic fluctuation of demand, traditional generation, and renewable generation in the electric grid. Load demands in each load bus are varied within 60% to 140% of their default value with uniform probability. Active power generation by traditional generators is varied according to normal distributions with means equal to the generator outputs obtained by running the PyPower DC-OPF function in the default setting and the standard deviations equal to 50% of the corresponding means. Reactive powers are varied according to uniform distributions in the range of minimum and maximum reactive power output of the generators. Total renewable generation in the system is considered to be varied according to a normal distribution with a mean equal to 10% of active generation in the system and a standard deviation equal to 5% of the mean according to the current generation share of different sources reported by the U.S. Energy Information Administration (EIA) (80). Subsequently, this renewable generation is distributed among the randomly selected 20% of load buses of the grid.

Table 1: Overview of IEEE Test Systems (30)

Test System	No. of Buses	No. of Generators	No. of Branches
Case 9	9	3	9
Case 14	14	5	20
Case 24	24	33	38
Case 30	30	6	41
Case 118	118	54	186

5.2 EMF-GNN Training Process

The surrogate models, including the EMF-GNN, MF-GNN (29), and SF-GNN, are implemented using the PyTorch framework (81), with GNN architectures constructed through the PyTorch Geometric library (82). To ensure optimal performance, we fine-tune the hyperparameters of all models based on the validation dataset. The architectures of both the LF-GNN and HF-GNN components within the EMF-GNN, as well as the SF-GNN, maintain a consistent structure in their hidden layers but differ in their input and output layers. Each LF-GNN model consists of three TagConv GNN layers (67) with 256 hidden units each, using sum aggregation and ReLU activation functions, except in the output layer. The HF-GNN blocks consist of three k-GNN (66) layers with 64 hidden units and the same activation function setting. Training extends over 400 epochs, utilizing the Adam optimization algorithm (83) with a learning rate of 0.001, and is supported by an NVIDIA RTX 4090 GPU.

For the training of the EMF-GNN, we utilize twice the amount of LF data as HF data. Additionally, to maintain a fair comparison, the SF-GNN, which only utilizes HF data, was provided with 40% more HF data points. This adjustment compensates for the lesser computational demands of the DCPF low-fidelity simulations, which require approximately 20% of the computational resources needed for the ACPF high-fidelity simulations.

5.3 Evaluation Metrics

To assess the performance of surrogate models and the accuracy of the optimal solutions derived from surrogate-based OPF, we employ Mean Squared Error (MSE) and Mean Absolute Error (MAE). MSE is defined as $MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$, and MAE is defined as $MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$, where $\hat{y}_i$ and y_i represent the estimated and actual values for the i -th measurement, respectively. Additionally, the Mean Relative Error (MRE), defined as $MRE = \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right|$, is used to evaluate the relative deviation of the estimated values from the true values in evaluating the accuracy of surrogate-based HCC-OPF. For a broader analysis of solution quality across different operational scenarios, the distribution of the solutions' expected generation costs for 200 trials is visualized and compared. To quantify the similarity between the distributions generated by the surrogate-based OPF and the benchmark ACPF, we use Jensen-Shannon Divergence (JSD) (84).

5.4 Experimental Results

5.4.1 Performance of EMF-GNN

Figure 2 compares the performance of SF-GNN and the proposed EMF-GNN as a surrogate for power flow solver in terms of MSE loss of voltage magnitudes and phase angles estimation against various amounts of high-fidelity data used to train the models. The results indicate that the proposed EMF-GNN consistently outperforms the SF-GNN for any amount of high-fidelity training data. Moreover, it can be observed that, for both voltage magnitudes (Figure 2 (a)) and phase angles (Figure 2 (b)), there exist elbows corresponding to 6000 training samples suggesting insignificant improvement of model accuracy for increasing the amount of training data beyond this point. Therefore, 6000 samples of high-fidelity training data are considered for all the subsequent experiments, achieving an optimal balance between accuracy and computational efficiency.

Furthermore, Table 2 compares the performance of proposed EMF-GNN with MF-GNN in (29) and SF-GNN for various IEEE cases. It can be seen that the multi-fidelity GNN method substantially improves the performance over SF-GNN. It can also be seen that EMF-GNN further improves MF-GNN in estimating the voltage phasors in different test cases. This is because of the modified architecture of the multi-fidelity scheme that allows better exploitation of the low-fidelity estimations.

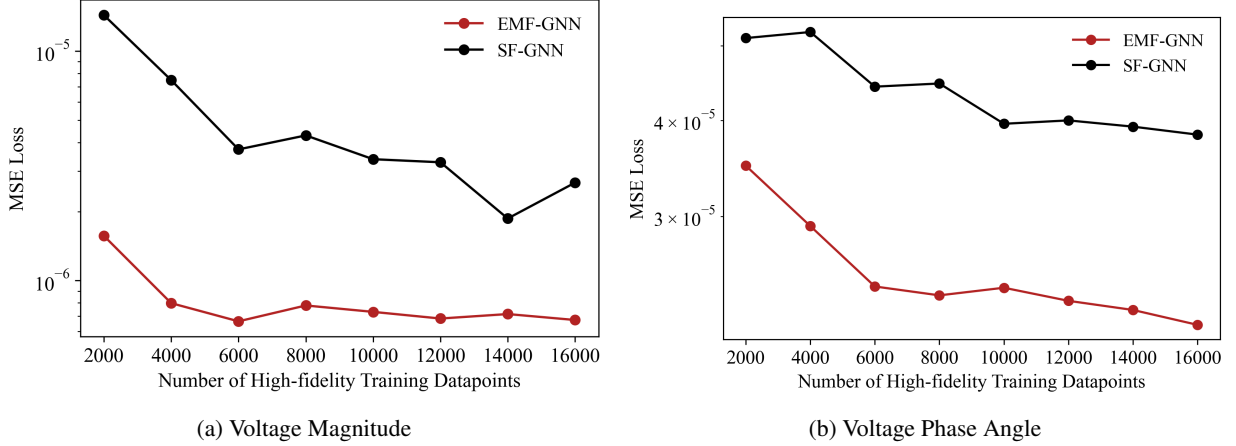


Figure 2: Comparison of mean square errors in voltage estimation for the IEEE 14-Bus system using multi-fidelity (EMF-GNN) and single-fidelity (SF-GNN) models. The x-axis shows the number of high-fidelity simulations for EMF-GNN, which also uses twice as many low-fidelity simulations. Considering high-fidelity simulations are five times costlier, SF-GNN, receives 40% more high-fidelity data points than EMF-GNN.

Table 2: Comparison of different metrics across different models and test cases without any contingencies in the grid. The values for voltage magnitude (V), and voltage phase angle (θ) are in p.u. and radian respectively.

Case	Metric	Measurement	SF-GNN	MF-GNN	EMF-GNN
Case 9	MSE	V	1.75E-05	8.00E-06	5.80E-06
		θ	1.03E-02	5.93E-03	5.61E-03
	MAE	V	2.69E-03	1.89E-03	1.34E-03
		θ	6.61E-02	4.59E-02	4.26E-02
Case 14	MSE	V	1.45E-06	1.01E-06	6.78E-07
		θ	3.90E-05	3.70E-05	2.47E-05
	MAE	V	8.42E-04	7.26E-04	5.79E-04
		θ	3.96E-03	3.68E-03	3.15E-03
Case 30	MSE	V	1.36E-05	1.15E-05	8.10E-06
		θ	1.53E-03	1.37E-03	1.30E-03
	MAE	V	2.56E-03	2.45E-02	1.87E-03
		θ	2.54E-02	2.40E-02	2.32E-02
Case 118	MSE	V	1.03E-02	7.40E-03	1.78E-03
		θ	8.37E-02	7.52E-02	7.83E-02
	MAE	V	1.47E-02	1.78E-02	1.32E-02
		θ	1.99E-01	1.84E-01	1.86E-01

5.4.2 Performance of HCC-OPF

The subsequent experiments assess the efficacy of the proposed HCC-OPF framework using the EMF-GNN models. HCC-OPF with a non-surrogate ACPF solver is used as the ground truth to evaluate the performance of HCC-OPF with the proposed surrogate EMF-GNN in estimating the optimal solution and the objective function in various load and renewable generation scenarios. Specifically, we evaluate the performance of HCC-OPF using 200 different trials. Each trial includes creating a new multi-variate Gaussian distribution of load and renewable generation at each bus. These multi-variate Gaussian distributions are created by using a perturbed value of the baseline loads for each IEEE case as the mean of distribution. An exponential decay function is used to estimate the correlation between load and generations at different buses. The total random renewable generation is set to be between 5% and 15% of total active generations by traditional generators, and assigned to 20% of randomly selected load buses. It is noteworthy that for line capacities, the values suggested by (16) are employed.

Figure 3 shows the distribution of optimal expected cost across the 200 trials using SF-GNN and EMF-GNN as surrogates within HCC-OPF. It is observed that, for all cases, HCC-OPF with EMF-GNN results in costs that are closer to optimal costs achieved by HCC-OPF when exact ACPF solver. The distribution similarities are further quantitatively assessed in Table 3 using JSD metric. Specifically, lower JSD values of EMF-GNN demonstrate less divergence from distributions obtained using exact ACPF. Additionally, Table 3 compares the relative error in the optimal objective functions, $\bar{C}$, and the absolute error in optimal generator outputs (decision variables) for HCC-OPF using both SF-GNN and EMF-GNN. The multi-fidelity model surpasses the single-fidelity model in these metrics, showcasing its enhanced capability in accurately converging to the optimal solution of the OPF under uncertainty. Additionally, it is evidenced that as the system size increases, the proposed surrogate models do not suffer from scalability problems and the error across test cases with different sizes remains stable.

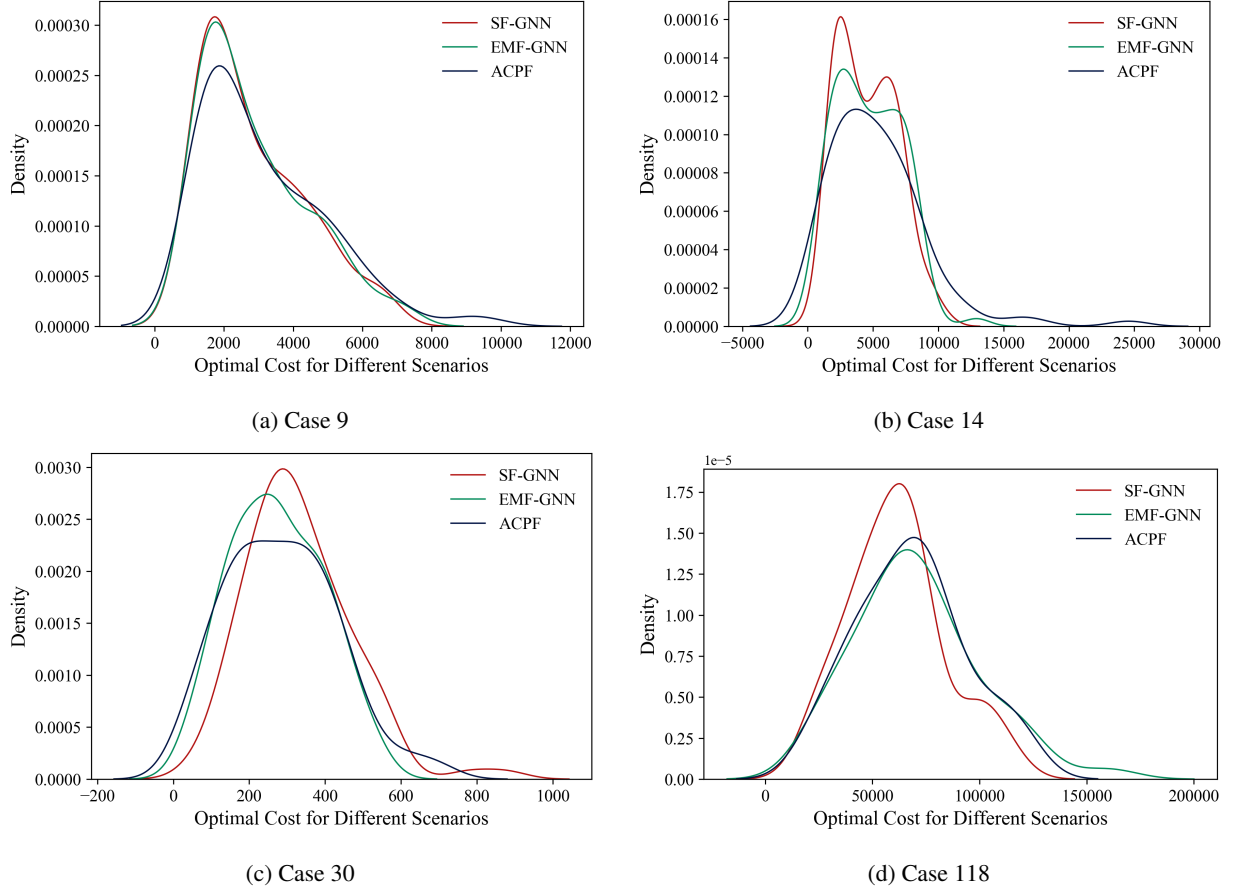


Figure 3: Comparison of the probability distributions of optimal expected costs across 200 trials over different test cases using HCC-OPF with exact ACPF solver, SF-GNN surrogate, and EMF-GNN surrogate.

Moreover, Figure 4 compares the mean computation time for each iteration and the whole optimization process across different IEEE test cases for using SF-GNN, EMF-GNN, and exact ACPF within the HCC-OPF framework. Expectedly, it can be seen that using surrogates can substantially reduce the computational cost of OPF under uncertainty. It must be noted that the proposed EMF-GNN surrogate only slightly increases the computational cost compared to the SF-GNN. Interestingly, the differences in the computational cost between the two methods decrease as the size of test cases becomes larger. This is attributed to the fact that the computational cost of processing the LF-GNN becomes less significant compared to the HF-GNN for larger graphs.

The results so far have shown the effectiveness of employing the proposed EMF-GNN within the proposed HCC-OPF framework. Next, we investigate the enhancement achieved by the proposed *hybrid* approach itself. Specifically, we compared the proposed HCC-OPF with surrogates-based CC-OPF, where the surrogates are fully trusted in indicating a constraint violation. Table 4 shows a comparison of different metrics that compare the solution achieved by these methods with the solution of CC-OPF using exact AC simulation for 200 trials. The results demonstrate significant enhancement achieved by the proposed hybrid approach compared to the surrogate-based CC-OPF. Notable

Table 3: Comparison of different optimization metrics over different test cases in handling operational constraints for SF-GNN, and EMF-GNN surrogates. The errors for the objective function and optimal generation are relative error and absolute error respectively.

Case	Surrogate	Performance Metric		
		Obj. Func. Rel. Error	Opt. Sol. Abs. Error	JSD
Case 9	SF-GNN	0.09	10.33	0.09
	EMF-GNN	0.08	9.81	0.076
Case 14	SF-GNN	0.18	7.08	0.16
	EMF-GNN	0.17	5.72	0.14
Case 30	SF-GNN	0.44	6.65	0.13
	EMF-GNN	0.17	3.68	0.1
Case 118	SF-GNN	0.1	9.21	0.03
	EMF-GNN	0.08	9.12	0.02

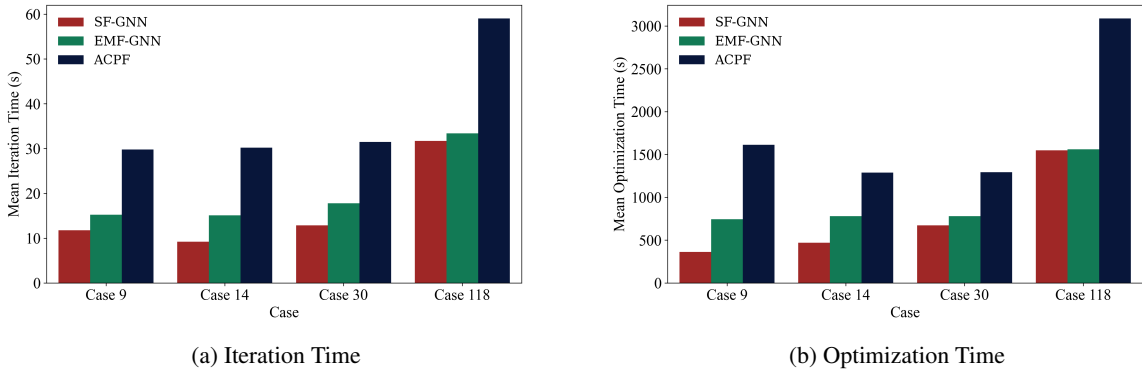


Figure 4: Comparison of computational times of HCC-OPF using exact ACPF solver, SF-GNN, and EMF-GNN.

improvements include a substantial reduction in constraint violation probability at the optimal solution, as well as solutions closer to the optimal solutions achieved when using exact ACPF. Expectedly, it can also be seen that the proposed HCC-OPF increases the computation time of each iteration. However, a minimal increase in overall optimization time is observed. This contrast is attributed to faster convergence of optimization in the hybrid method. These advancements highlight the computational efficiency of the proposed hybrid optimization method.

Table 4: Changes in different metrics for multi-fidelity surrogate-based HCC-OPF compared to multi-fidelity surrogate-based CC-OPF for Case 118.

Method	Constraint Violation Probability	Objective Function MRE	JSD	Iteration Time (s)	Optimization Time (s)
HCC-OPF using EMF-GNN	0.06	0.03	0.02	96.37	1635.21
CC-OPF using EMF-GNN	0.23	0.11	0.08	33.02	1554.21

5.4.3 Performance of HCC-OPF under N-1 Contingency

Finally, the adaptability of the proposed HCC-OPF with EMF-GNN surrogate has been assessed under N-1 security constraints or contingencies, characterized by random failures of a single traditional generator or transmission line in the grid. Figure 5 shows the distribution of expected cost for the solution of the 200 trials with N-1 contingencies using SF-GNN and EMF-GNN as surrogates within HCC-OPF. It is observed that, even with N-1 contingencies, the GNN

surrogates are able to generate solutions close to those achieved when using an exact ACPF solver. It is also observed that EMF-GNN outperforms SF-GNN. This outperformance is detailed in Table 5, which summarizes both the JSD distribution similarity and error of the objective function values obtained by HCC-OPF using MF-GNN and SF-GNN. It is notable that the errors associated with the multi-fidelity models do not significantly increase compared to the results presented in 3 even though the surrogate models face random topology changes in the input data in this experiment, highlighting the robustness of EMF-GNN when facing small changes in the grid topology.

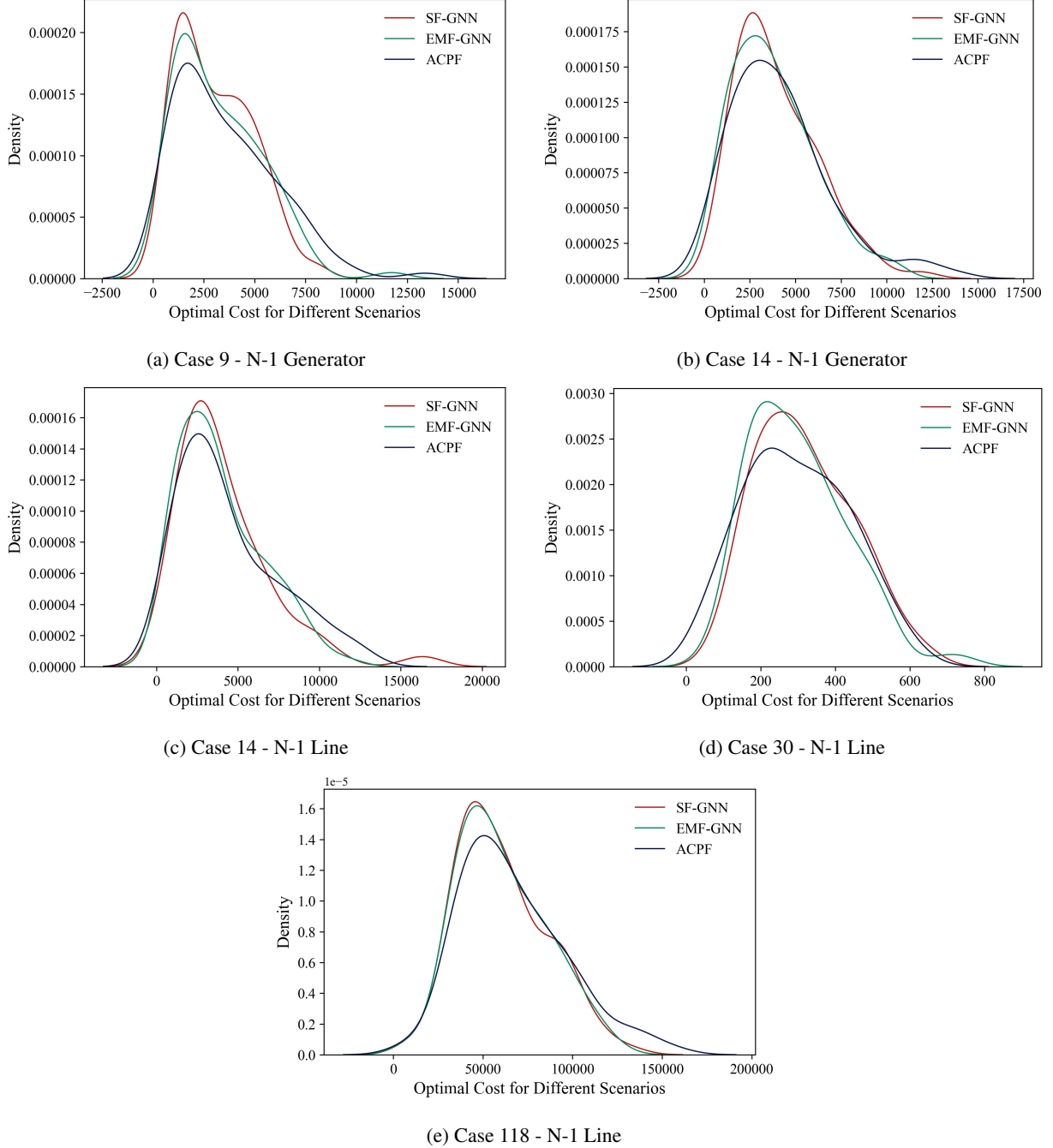


Figure 5: Comparison of optimal cost distribution functions for different scenarios over different test cases with N-1 security constraint contingencies using Newton Raphson method, and single-fidelity and multi-fidelity models.

Table 5: Comparison of different optimization metrics across different surrogate and test cases in handling N-1 constraints. The errors for the objective function and optimal generation are relative error and absolute error respectively.

Constraint	Case	Surrogate	Performance Metric	
			Obj. Func. MRE	JSD
N-1 Generator	Case 9	SF-GNN	0.12	0.126
		EMF-GNN	0.11	0.088
	Case 14	SF-GNN	0.24	0.07
		EMF-GNN	0.2	0.08
N-1 Line	Case 14	SF-GNN	0.24	0.11
		EMF-GNN	0.15	0.09
	Case 30	SF-GNN	0.27	0.063
		EMF-GNN	0.18	0.068
	Case 118	SF-GNN	0.103	0.1
		EMF-GNN	0.097	0.09

6 Conclusion

Modern power systems are facing increasing uncertainty on both load and generation sides due to the integration of renewable energy sources and the rising demand for electrification. Integrating these uncertainties into the OPF problem will lead to a significant computational burden. Surrogate models present a viable solution for faster analysis, but advanced surrogate models are typically data-hungry and expensive to train. Moreover, surrogate-based optimization can lead to unreliable solutions due to erroneous constraint violations. To address these challenges, we propose an innovative surrogate-based HCC-OPF methodology with EMF-GNN models as power flow solver surrogates. On the surrogate side, the proposed EMF-GNN leverages the explicit representation of grid topology and integrates low-fidelity data with high-fidelity simulations to reduce training costs and computational demands significantly. On the optimization side, the proposed hybrid approach ensures maximum accuracy and minimizes errors in constraint violations by selectively employing the AC power flow solver when the surrogate estimations approach critical thresholds. The experiments in this work validate the proposed methodology under different system sizes and conditions and comprehensive load and renewable generation uncertainty. Specifically, the results demonstrated the outperformance of the EMF-GNN model compared to single-fidelity and previously developed multi-fidelity GNN models. Additionally, the experiments show that the proposed HCC-OPF method outperforms conventional CC-OPF and provides accurate estimations of optimal solutions under high-dimensional uncertainty for normal conditions and N-1 security contingencies. Future research could explore more adaptive hybrid approaches to further enhance the performance of HCC-OPF and investigate heterogeneous GNN models to refine the accuracy of surrogate models.

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